

A hybrid multi-stage decision-making method with probabilistic interval-valued hesitant fuzzy set for 3D printed composite material selection

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ABSTRACT

The 3D printed composite material selection is of great interest due to its extensive application prospect and can be considered as a challenging multiple-criteria decision making (MCDM) issue. The hesitation and uncertainty of experts are difficult to measure, and the high degree of interaction among criteria is often overlooked in the decision process. In addition, composites are required to serve in harsh environments for various mechanical and industrial fields, resulting in the degradation of mechanical properties. In this study, a hybrid multi-stage decision-making method is developed to conduct 3D printed composite material selection in harsh environments. The theory of probabilistic interval-valued hesitant fuzzy set (PIVHFS) is proposed to characterize the decision-making information of experts, which can effectively quantify the assessment in uncertain environments. An integrated method that combines Choquet fuzzy integral and Shapley value is proposed to obtain the weight vector of criteria, which can reflect the mutual influence between criteria and their overall importance. The final decision-making result and the optimal alternative can be calculated by the PIVHFS-based Tomada de Decisão Interativa Multicritério and Technique for order preference by similarity to an ideal solution (TODIM-TOPSIS) method. An empirical application, i.e., 3D printed composites in the background of automotive chassis, is applied to validate the application of the proposed method. Comparative analysis, sensitivity analysis, and managerial implications are also conducted to illustrate the validity of the method. This paper provides a valuable tool for addressing the material selection issue of 3D printed composites from a multi-criteria perspective.

1. Introduction

The rapid development of advanced manufacturing accelerates social and economic progress and greatly improves productivity, economic profitability, and product competitiveness (Nazir et al., 2023). 3D printing (additive manufacturing), as an emerging manufacturing technology, has drawn great attention in both scientific and industrial research (Darwish et al., 2021). Based on the digital information, 3D printing manufactures products by heating composites to a molten state and printing the fused composites layer by layer on the plate. Compared with conventional manufacturing of composites, 3D printed composites have advantages in terms of lightweight, energy-saving, and environmental protection (Cheng et al., 2023; Li et al., 2023; Zhou et al., 2023). In addition, 3D printing technology and its products have been widely applied in aerospace, transportation, and biomedical

fields, bringing huge social and economic benefits. With the growing competitiveness of advanced composites, 3D printing composites have become a vital channel to achieve green production with low energy consumption and high economic efficiency (Divakaran et al., 2022; Tafakkori et al., 2022).

In comparison with conventional composites, 3D printed composite materials are designed based on multiple parameters, which mainly include printing strategies and material types, with a high degree of designability. The rich design parameters of 3D printed composite materials also provide the possibility to achieve higher mechanical properties. Studies have shown that printing strategies and materials, e.g., printing matrix, reinforced fibers, printing directions, and stacking sequences, have appreciable influences on the mechanical performances of 3D printed composites (Wang et al., 2020a, 2022a, 2020b; Zhu et al., 2023). Hence, multiple 3D printed composites with

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different mechanical properties can be designed to satisfy diverse application needs. For specific engineering contexts, the material selection of 3D printed composites is significant for the design and production of composite products. In engineering applications, composite materials are required to work in harsh environments, e.g., hygrothermal aging environment. Many scholars have found that the hygrothermal aging environment has a complex mechanism to influence the mechanical properties of composites (Kallannavar and Kattimani, 2023; Pizzorni and Prato, 2023; Wang et al., 2021). The capacity of composite materials to sustain harsh environments is increasingly becoming an essential material selection reference. In addition, with the growing problems of environmental deterioration and the urgent demand for social and economic development, the environmental, social and economic benefits of 3D printed composite materials need also be considered in the material selection process. Therefore, the material selection of 3D printed composites considering mechanical properties, economy, environment, society, and stability for harsh environments is critical for the product design of 3D printed composites (Chen et al., 2019; Wang et al., 2022b; Zavadskas et al., 2017). Selecting the optimal alternative among numerous materials usually requires compromises between competing multiple criteria, which is a complicated multiple-criteria decision making (MCDM) issue. To address this issue, it is meaningful and necessary to propose a valuable MCDM method to obtain the composite material alternative with optimal overall performance and satisfy the practical requirements.

This study aims to propose a multi-stage MCDM method for the material selection of 3D printed composites in hygrothermal aging environments. To better depict the hesitancy and inaccurate judgment of decision-makers, the probabilistic interval-valued hesitant fuzzy set (PIVHFS) is presented. The PIVHFS-based fuzzy integral and Shapley value are applied to explore the effects of interaction among criteria. In addition, based on PIVHFS, the TOPSIS method incorporating the TODIM method is introduced to consider the correlation and dominance between criteria and further rank the sort of alternatives. An evaluation criterion model considering multiple factors, which includes economic, environmental, social, mechanical, and stability aspects, is proposed to satisfy the design and application requirements of 3D printed composites. As a case study, the 3D printed continuous glass fiber reinforced composites with the background of automotive chassis are introduced to validate the proposed decision-making method. This study provides a valuable research reference for the material selection issue on 3D printed composite material. The proposed multi-stage MCDM considering criterion interaction effects and criterion correlation effects is of great value to explore the criterion effects on the final decision results.

The remaining main points of this paper are organized as follows. The literature reviews of the material selection issue, fuzzy theory, and MCDM approach for material selection are conducted in Section 2. Section 3 proposes the concept of PIVHFS, fuzzy integral, and Shapley value. Section 4 introduces the multi-stage decision-making method. In Section 5, an illustrative application, that is the automotive chassis, considering the material investigation of 3D printed composites and multiple criteria is introduced to select the optimal alternative. Section 6 introduces the comparative analysis, sensitivity analysis, managerial implications, and research limitations. Finally, Section 7 concludes the findings of the study.

2. Literature review

2.1. Literature review on the material selection issue

As a valuable decision support tool, material selection is of great significance to designing composite material products and helps designers to make the optimal decision. Numerous types of research on material selection have been conducted to meet the diverse requirements of multiple fields. Haq et al. developed an extended MAIRCA decision

method to address the sustainable material selection of a human-powered aircraft wing-spar considering the economic, mechanical, and sustainable criteria (Ul Haq et al., 2023). Yoris-Nobile et al. designed a kind of 3D printed artificial reef and further solved its material selection problem by considering environment, economy, and bio receptivity aspects (Yoris-Nobile et al., 2023). Kazemian et al. attempted to design intraoral stents for patients with head and neck cancer to implement better radiation therapy. The mechanical and physical properties are mainly considered to address its material selection problems (Kazemian et al., 2023). Zhang et al. proposed a comprehensive MCDM method to solve the material selection issue of 3D printed composites reinforced by continuous carbon fibers considering the criteria of mechanics, society, environment, and economy (Zhang et al., 2020a). Wang et al. further established a multiple criteria model for the material selection of 3D printed polyamide composites, which took into account the mechanical properties at multiple strain rates along with social, environmental, and economic benefits (Wang et al., 2022b).

The literature reviews illustrate that with the rapid development of advanced composite materials, many fields are facing the challenge of material selection. 3D printed composites have more prominent material selection issues due to the characteristics of multiple design parameters. The multiple criterion model, as a tool to depict material performances of various aspects, is commonly viewed as a vital link for addressing material selection issues (Khoshnava et al., 2018). In current studies, the criteria of environment, economy, society, and mechanics are commonly cited to depict the overall performances of multiple materials (Wang et al., 2022b; Zhang et al., 2020a). The current considered mechanical criteria mainly stay in the freshly manufactured composites. However, in practical applications, harsh environments, one as the hygrothermal aging environment, have a huge impact on the properties of composite materials in service, which cannot be reflected by the freshly manufactured composites (Wang et al., 2021). To further consider the hygrothermal aging effects on composite material mechanical properties, it is essential to establish a multiple-criterion model, which takes into account the stability criterion to the hygrothermal aging environment.

2.2. Literature review on the fuzzy theory for material selection

Material selection usually involves many alternatives, which requires experts to evaluate each alternative in uncertain decision-making contexts (Li et al., 2021). Due to the different individual perspectives and competence of decision-makers from various fields, the alternative evaluation process may produce inaccurate and uncertain information, resulting in inappropriate decision results (Deliktaş and Şahinöz, 2023). Fuzzy sets theories proposed by Zadeh have been proved as excellent solutions for expressing decision information in uncertain contexts and conducted by numerous studies to address uncertain problems in material selection (Zadeh, 1996). Zhang et al. introduced the Pythagorean fuzzy set and Pythagorean fuzzy weighted average operator to solve the uncertainty problems in the material selection process (Zhang et al., 2020b). Farid and Riaz proposed new aggregation operators based on single-valued neutrosophic sets to solve the material selection problems of cryogenic storage containers (Farid and Riaz, 2022). Haq et al. proposed an MCDM method combining the Interval-Valued Neutrosophic Sets and MAIRCA method to simultaneously process the subjective indicators with uncertain information and the objective indicators with crisp information (Ul Haq et al., 2023). To address the description issues of experiment errors in material selection along with the fuzzy problems, the triangle fuzzy set was suggested by Zhang et al. and Wang et al. (Wang et al., 2022b; Zhang et al., 2020a). In addition, some other fuzzy theories were also conducted to settle the uncertain and fuzzy problems, e.g., intuitionistic fuzzy set (Chen, 2023), interval-valued fuzzy sets (Gohain et al., 2023), and interval type-2 fuzzy set (Altay et al., 2023).

The literature illustrates that current material selection studies mainly focus on solving the uncertain problems caused by a single fuzzy

context, lacking the means to handle comprehensive fuzzy problems. In the material selection process, decision information commonly meets the matters of incomplete divergence and inaccurate judgment, and the hesitancy of information from decision-makers is neglected, which makes the material selection a complex fuzzy issue. The hesitant fuzzy set (HFS), proposed by Torra, has been viewed as a useful tool to solve hesitant information of decision-makers (Torra, 2010). However, due to the lack of accurate and clear judgments, decision-makers cannot express their opinions by several specific numbers in the way of HFS (Zhang et al., 2019). Driven by this problem, some researchers proposed to replace the specific numbers with interval-valued numbers and further defined the concepts of interval-valued hesitant fuzzy sets (IVHFS) (Wang and Wang, 2014). In addition, the decision information of decision-makers is not fully divergent, which makes IVHFS invalid sometimes (Divsalar et al., 2022). To address this matter, Song et al. introduced the probabilistic information to consider certain convergence of decision-makers, and further proposed the concept of probabilistic interval-valued hesitant fuzzy sets (PIVHFS) (Song et al., 2019). PIVHFS has demonstrated its superiority in solving uncertain problems, e.g., the hesitancy of information, incomplete divergence, and inaccurate judgment. Moreover, PIVHFS shows the potential for resolving fuzzy problems in material selection, which has not yet been explored.

2.3. Literature review on the mcdm approach for material selection

In the MCDM process, the criterion weights represent the importance of each evaluation criterion, Which is viewed as a vital link to making logical decisions. Some criterion weight methods have been conducted in the material selection process to obtain the proper weights, e.g., the AHP method (Tian et al., 2018), entropy method (Guo et al., 2022), best-worst method (BWM) (Zhang et al., 2020a), QFD method (Chen et al., 2019), DEMATEL method (Wang et al., 2022b) et al. These methods attempt to compute the criterion weights from multiple perspectives. However, the interactive characteristics and interdependencies in criteria are commonly neglected. In the actual decision process, criteria may operate on other criteria to indirectly affect the criterion weights in addition to having an impact on the criterion weights directly, which implies that there are interactions among criteria. Fuzzy measure and fuzzy integral have been proved as effective tools to measure the interaction among elements (Wei et al., 2017). On basis of the fuzzy measure and fuzzy integral theory, Shapley value is originally proposed by Shapley to solve the problems in game theory (Divsalar et al., 2022). Recently, the Shapley value has been explored by researchers to measure the importance of each or several criteria in the MCDM process (Zhang et al., 2017). Compared with conventional weight methods, the Shapley value can extract the effects of interactions among criteria and further obtain the overall importance weights (Tan et al., 2015). In the manufacturing process of 3D printed composites, there exist complex interactions among multiple criteria. For instance, in the selection process of 3D printed composites, the criterion “material cost” will affect “maintenance cost”, and “recycling potential”. In addition, the “material cost” is also affected by “raw material extraction” “lightweight level”, and “operational life”. However, there remain large gaps in the research of the criterion interactions for 3D printing composite material selection. Therefore, it is relevant to solve the material selection of 3D printing composites with the Shapley value method. In addition, the application of Shapley value in combination with PIVHFS is meaningful for solving the criterion interaction effects in uncertain and hesitant contexts, Which also provides references for the application of complex fuzzy sets with Shapley value.

Numerous studies have been conducted on MCDM methods to address material selection issues. Raigar et al. introduced the Proximity Indexed Value (PIV) method and the Best Worst Method (BWM) to rank the processes of additively manufactured composites (Raigar

et al., 2020). Zhang et al. proposed the triangular fuzzy number (TFN) based Grey Relational Analysis method and VlseKriterijumska Optimizes I Compromising Resented (G-VIKOR) method to obtain the optimal printing strategy of 3D printed continuous carbon fiber-reinforced composites (Zhang et al., 2020a). In addition, the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) (Huang et al., 2009; Memari et al., 2019), Tomada de Decisão Interativa Multicritério (TODIM) (Liu et al., 2021; Zindani et al., 2017), step-wise weight assessment ratio analysis (SWARA) (Ruzgys et al., 2014), VlseKriterijumska Optimizes I Compromising Resented (VIKOR) (Peng et al., 2023a,b), Complex Proportional Assessment (COPRAS), and data envelopment analysis (DEA) were also applied to material selection (Mousavi-Nasab and Sotoudeh-Anvari, 2017).

Among the methods, the TOPSIS method, proposed by Hwang and Yoon, is an acclaimed method to address the MCDM problem, which can be applied to a large number of criteria and alternatives (Hwang and Yoon, 1981). Its main idea is to compute the relative distance of each alternative to the negative ideal solution (NIS) and positive ideal solution (PIS) and further rank the alternatives by the largest distance to NIS and the closest distance to PIS (Celikbilek and Tuysuz, 2020; Hwang and Yoon, 1981). TOPSIS method has been successfully applied in many fields, e.g., material selection, low-carbon port evaluation, green supplier selection, bank evaluation, and nuclear risk management et al. exhibiting huge application value (Awodi et al., 2023; Corrente and Tasiou, 2023; Hajiaghahi-Keshteli et al., 2023; Yang et al., 2022). However, several limitations remain. Studies show that TOPSIS has difficulty considering the correlation and dominance between criteria (Celikbilek and Tuysuz, 2020; Kuo, 2017; Wang and Wang, 2014). How to settle these problems becomes the key to advance further research and application of the TOPSIS method.

The TODIM method, proposed on the basis of prospect theory, can effectively address the bounded rational problem of decision-makers (Gomes and Lima, 1992; Kahneman and Tversky, 1979). Its main idea is to establish the relative dominance degree function of the alternative relative to other alternatives and further make decisions (Xu et al., 2022; Zindani et al., 2017). The studies of Wang et al. and Arya et al. explored transforming the dominance degree of alternatives to the degrees of criteria, which provided the possibility to construct a bridge for the correlation and dominance between criteria (Arya and Kumar, 2020; Wang et al., 2022b). Therefore, there exists a potential for the TODIM method to address the limitations of the TOPSIS approach. It is relevant to explore a multi-stage decision-making method incorporating the TOPSIS method and TODIM method and its application on material selection.

2.4. Research limitations and contributions

In light of the literature review, some limitations can be concluded: (1) the current research on material selection mainly concentrates on the performances of freshly manufactured composites. No research has considered the effects of hygrothermal aging environments during service on material selection; (2) in the material selection process, many fuzzy theories have been conducted to solve single uncertain issues. However, there exists little literature researching on complex fuzzy environments, which consider hesitancy information, information incomplete divergence, and inaccurate judgment, of 3D printed composites; (3) in the practical decision process of material selection, the interactions among criteria, which have a certain influence on the criterion weights, are commonly neglected; (4) many MCDM issues have been successfully settled by TOPSIS methods combining multiple fuzzy theory. However, little research has been explored to address the correlation and dominance of criteria in the TOPSIS method.

For filling the abovementioned knowledge gaps, this study proposes a hybrid multi-stage decision-making method with PIVHFS to settle the 3D printed composite material selection problem. Compared with previous studies, four distinctive contributions are summarized

as follows: (1) the theory of PIVHFS is designed to characterize the decision-making information of experts, which can effectively quantify the assessment in hesitant and uncertain environments; (2) a hybrid multi-stage decision-making method employing PIVHFS-based Shapley value to settle the criterion interactions and the PIVHFS-based TODIM-TOPSIS method to consider the criterion correlation and dominance is proposed to address the 3D printed composite material selection in the hygrothermal aging environment; (3) a hierarchical multiple criterion model comprehensively considering the stability for the hygrothermal aging environment along with mechanical, economic, environmental, and social aspects is established. In addition, a decision matrix construction method incorporating mechanics experiments and expert assessments is proposed; (4) an illustrative application, that is the 3D printed continuous glass fiber reinforced composites with the background of automotive chassis, is introduced to validate the proposed decision-making method.

3. Preliminaries

3.1. Probabilistic interval-valued hesitant fuzzy set

The theory of PIVHFS is defined based on the hesitant fuzzy set, interval number, and probabilistic information, which can effectively address the issues of hesitancy information, incomplete divergence, and inaccurate judgment. The relevant definitions are shown as follows.

3.1.1. The concept of PIVHFS

Definition 1. (Torra, 2010) For a reference set X , the function $\tilde{h}_D(x)$ defines an HFS D on X and returns a finite subset of $[0,1]$.

$$D = \{x, \tilde{h}_D(x) | x \in X\} \quad (1)$$

where $\tilde{h}_D(x)$, a set of certain values in $[0,1]$, denotes the possible membership degree of elements x to set D . $\tilde{h}_D(x)$ is a fundamental unit of the HFS, which is named a hesitant fuzzy element (HFE).

Definition 2. (Wang and Wang, 2014; Xu and Da, 2002) An interval number $\tilde{a}$ satisfies $\tilde{a} = [a^L, a^U] = \{x | a^L \leq x \leq a^U\}$, which is represented by Eq. (2).

$$\tilde{a} = [a^L, a^U] \quad (2)$$

where a^L is the lower limit of $\tilde{a}$ and a^U represents the upper limit.

If $a^L \geq 0$, the interval number $\tilde{a}$ transformed into a positive interval number. $\tilde{a} = [a^L, a^U]$ and $\tilde{b} = [b^L, b^U]$ are two positive interval numbers, the operations of which can be defined by Eq. (3).

$$\begin{aligned} \tilde{a} + \tilde{b} &= [a^L + b^L, a^U + b^U]; \\ \tilde{a}^\eta &= [(a^L)^\eta, (a^U)^\eta]; \\ \eta \tilde{a} &= [\eta a^L, \eta a^U]; \\ \tilde{a} \cdot \tilde{b} &= [a^L \cdot b^L, a^U \cdot b^U]; \\ \tilde{a} \cup \tilde{b} &= [\max\{a^L, b^L\}, \max\{a^U, b^U\}]; \\ \tilde{a} \cap \tilde{b} &= [\min\{a^L, b^L\}, \min\{a^U, b^U\}]. \end{aligned} \quad (3)$$

where η represents a real number satisfying, $\eta \in [0, 1]$.

If $\tilde{a}_1 = [a_1^L, a_1^U]$ and $\tilde{a}_2 = [a_2^L, a_2^U]$ are two interval numbers. The length of $\tilde{a}_1$ and $\tilde{a}_2$ is presented as $len(\tilde{a}_1) = a_1^U - a_1^L$ and $len(\tilde{a}_2) = a_2^U - a_2^L$. Then, the possible degree of $\tilde{a}_1 \geq \tilde{a}_2$ and $\tilde{a}_1 \leq \tilde{a}_2$ can be defined by Eqs. (4) and (5):

$$p(\tilde{a}_1 \geq \tilde{a}_2) = \max \left\{ 1 - \max \left\{ \frac{a_2^U - a_1^L}{len(\tilde{a}_1) + len(\tilde{a}_2)}, 0 \right\}, 0 \right\} \quad (4)$$

$$p(\tilde{a}_2 \geq \tilde{a}_1) = \max \left\{ 1 - \max \left\{ \frac{a_1^U - a_2^L}{len(\tilde{a}_1) + len(\tilde{a}_2)}, 0 \right\}, 0 \right\} \quad (5)$$

Definition 3. (Wang and Wang, 2014) For a reference set X , an interval-valued hesitant fuzzy set (IVHFS) D on X can be defined as follows:

$$E = \{x, \tilde{h}_E(x) | x \in X\} \quad (6)$$

$$\tilde{h}_E(x_i) = \{\gamma | \gamma \in \tilde{h}_E(x_i)\} \quad (7)$$

where $\tilde{h}_E(x)$ is an interval-valued hesitant fuzzy element (IVHFEs) with the values in $[0,1]$; $\gamma = [\gamma^L, \gamma^U]$ represents an interval number. If the upper and lower limits of the interval values are the same, IVHFS changes into HFS.

Definition 4. (Zhang et al., 2019) For a reference set X , a PIVHFS D on X is defined by function Eq. (8), which returns a finite subset of $[0,1]$ when applied to X . Probabilistic interval-valued hesitant fuzzy element (PIVHFE) is a fundamental unit of the PIVHFS, which is represented by Eq. (8).

$$D = \{x, \tilde{h}_D(p_x) | x \in X\} \quad (8)$$

where $\tilde{h}_D$ are the membership degrees of x to set D , which belong to $[0,1]$. $\tilde{h}(p)$ is used to represent the PIVHFE, which can be expressed by Eq. (9).

$$\tilde{h}(p) = \{\tilde{\gamma}_l(p_l) | l = 1, 2, \dots, |\tilde{h}(p)|\} \quad (9)$$

where $\tilde{\gamma}_l = [\gamma_l^L, \gamma_l^U]$ is an interval number, which denotes the membership degree. The γ_l^L is the lower limit of $\tilde{\gamma}$ and γ_l^U represents the upper one. p_l is the probability of $\tilde{\gamma}_l$, $\sum_{l=1}^{|\tilde{h}(p)|} p_l < 1$. The $|\tilde{h}(p)|$ is the number of membership degrees.

3.1.2. Operations of PIVHFE

The PIVHFE must be processed before performing basic operations. Based on the proposed normalization rules for PHFEs, Zhang and Xu further proposed the normalization rules for PIVHFEs (Zhang et al., 2019; Zhou and Xu, 2018). Definition 5 and Define 6 are two basic normalization rules.

Definition 5. (Zhang et al., 2019) $\tilde{h}(p)$ is a PIVHFE, $\tilde{h}(p) = \{\tilde{\gamma}_l(p_l) | l = 1, 2, \dots, |\tilde{h}(p)|\}$. If there exists $\sum_{l=1}^{|\tilde{h}(p)|} p_l < 1$, the loss of probabilistic information needs to be completed. A normalized PIVHFE $\bar{\tilde{h}}(p)$ is proposed to estimate the lost probabilistic information, which is expressed as follows.

$$\bar{\tilde{h}}(p) = \{\bar{\tilde{\gamma}}_l(\bar{p}_l) | l = 1, 2, \dots, |\bar{\tilde{h}}(p)|\} \quad (10)$$

where $\sum_{l=1}^{|\bar{\tilde{h}}(p)|} \bar{p}_l = 1$ and $\bar{p} = p_l / \sum_{l=1}^{|\bar{\tilde{h}}(p)|} p_l$, $l = 1, 2, \dots, |\bar{\tilde{h}}(p)|$.

when $\sum_{l=1}^{|\tilde{h}(p)|} p_l = 1$, $\tilde{h}(p) = \bar{\tilde{h}}(p)$. If every interval value has the same probability, the PIVHFEs can be transformed into IVHFEs, and further, when the membership degrees are represented by single numbers, the IVHFEs can be transformed into HFEs.

In most cases, the length of PIVHFEs is different from each other, which means PIVHFEs have a different number of membership degrees, making a difficult calculating problem. Thus, the length of each PIVHFE must be equal, which can be solved by Definition 6.

Definition 6. (Zhang et al., 2019) Two different PIVHFEs $\tilde{h}_1(p) = \{\tilde{\gamma}_i(p_i) | i = 1, 2, \dots, |\tilde{h}_1(p)|\}$ and $\tilde{h}_2(p) = \{\tilde{\gamma}_j(p_j) | j = 1, 2, \dots, |\tilde{h}_2(p)|\}$ are designed. Assume that $|\tilde{h}_1(p)| > |\tilde{h}_2(p)|$, the missing numbers of memberships in $|\tilde{h}_2(p)|$ need to be completed to be consistent with $|\tilde{h}_1(p)|$. The number of completed terms is expressed as Eq. (11).

$$|\tilde{\gamma}_m(p_m)| = |\tilde{h}_1(p)| - |\tilde{h}_2(p)| \quad (11)$$

where, $m = |\tilde{h}_2(p)| + 1, |\tilde{h}_2(p)| + 2, \dots, |\tilde{h}_1(p)|$, $\gamma_m = \min(\gamma_{2_i})$ and $p_m = 0$.

Definition 7. Three PIVHFs $\tilde{h}(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[\tilde{\gamma}_i^L, \tilde{\gamma}_i^U](p_i)\}$, $\tilde{h}_1(p) = U_{\tilde{\gamma}_i \in \tilde{h}_1}\{[\tilde{\gamma}_i^L, \tilde{\gamma}_i^U](p_{1_i})\}$ and $\tilde{h}_2(p) = U_{\tilde{\gamma}_j \in \tilde{h}_2}\{[\tilde{\gamma}_j^L, \tilde{\gamma}_j^U](p_{2_j})\}$ are designed. Eqs. (12)–(18) defines the operations of PIVHFs:

$$\tilde{h}^c(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[1 - \gamma_i^L, 1 - \gamma_i^U](p_i)\} \quad (12)$$

$$\tilde{h}^\lambda(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[(\gamma_i^L)^\lambda, (\gamma_i^U)^\lambda](p_i)\}, \lambda > 0 \quad (13)$$

$$\lambda \tilde{h}(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[1 - (1 - \gamma_i^L)^\lambda, 1 - (1 - \gamma_i^U)^\lambda](p_i)\}, \lambda > 0 \quad (14)$$

$$\tilde{h}_1(p) \cup \tilde{h}_2(p) = U_{\tilde{\gamma}_i \in \tilde{h}_1, \tilde{\gamma}_j \in \tilde{h}_2}\{[\max(\gamma_i^L, \gamma_j^L), \max(\gamma_i^U, \gamma_j^U)](p_i \cdot p_j)\} \quad (15)$$

$$\tilde{h}_1(p) \cap \tilde{h}_2(p) = U_{\tilde{\gamma}_i \in \tilde{h}_1, \tilde{\gamma}_j \in \tilde{h}_2}\{[\min(\gamma_i^L, \gamma_j^L), \min(\gamma_i^U, \gamma_j^U)](p_i \cdot p_j)\} \quad (16)$$

$$\tilde{h}_1 \oplus \tilde{h}_2 = U_{\tilde{\gamma}_i \in \tilde{h}_1, \tilde{\gamma}_j \in \tilde{h}_2}\{[\gamma_i^L + \gamma_j^L - \gamma_i^L \gamma_j^L, \gamma_i^U + \gamma_j^U - \gamma_i^U \gamma_j^U](p_i \cdot p_j)\} \quad (17)$$

$$\tilde{h}_1 \otimes \tilde{h}_2 = U_{\tilde{\gamma}_i \in \tilde{h}_1, \tilde{\gamma}_j \in \tilde{h}_2}\{[\gamma_i^L \gamma_j^L, \gamma_i^U \gamma_j^U](p_i \cdot p_j)\} \quad (18)$$

Definition 8. The score function of $\tilde{h}(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[\tilde{\gamma}_i^L, \tilde{\gamma}_i^U](p_i)\}$ can be expressed as follows:

$$s(\tilde{h}(p)) = [\sum_{i=1}^n \gamma_i^L p_i, \sum_{i=1}^n \gamma_i^U p_i] \quad (19)$$

where, $s(\tilde{h}(p))$ represents an interval value in $[0,1]$, which is denoted as s .

Definition 9. The standard deviation degree of $\tilde{h}(p) = U_{\tilde{\gamma}_i \in \tilde{h}}\{[\tilde{\gamma}_i^L, \tilde{\gamma}_i^U](p_i)\}$ can be defined as follows:

$$dv(\tilde{h}(p)) = [\sum_{i=1}^n p_i(\gamma_i^L - \sum_{i=1}^n \gamma_i^L \cdot p_i), \sum_{i=1}^n p_i(\gamma_i^U - \sum_{i=1}^n \gamma_i^U \cdot p_i)] \quad (20)$$

where $dv(\tilde{h}(p))$ represents an interval value, which is denoted as dv .

For any two PIVHFs $\tilde{h}_1(p)$ and $\tilde{h}_2(p)$, if $s(\tilde{h}_1(p)) > s(\tilde{h}_2(p))$, then $\tilde{h}_1(p) > \tilde{h}_2(p)$. If $s(\tilde{h}_1(p)) = s(\tilde{h}_2(p))$ and $dv(\tilde{h}_1(p)) > dv(\tilde{h}_2(p))$, then $\tilde{h}_1(p) < \tilde{h}_2(p)$. $\tilde{h}_1(p) = \tilde{h}_2(p)$ when $s(\tilde{h}_1(p)) = s(\tilde{h}_2(p))$ and $dv(\tilde{h}_1(p)) = dv(\tilde{h}_2(p))$.

Definition 10. The distance measure between PIVHFs $\tilde{h}_1(p) = U_{\tilde{\gamma}_i \in \tilde{h}_1}\{[\tilde{\gamma}_i^L, \tilde{\gamma}_i^U](p_i)\}$ and $\tilde{h}_2(p) = U_{\tilde{\gamma}_j \in \tilde{h}_2}\{[\tilde{\gamma}_j^L, \tilde{\gamma}_j^U](p_j)\}$ can be obtained by Eq. (21).

$$d(\tilde{h}_1(p), \tilde{h}_2(p)) = |s(\tilde{h}_1(p)) - s(\tilde{h}_2(p))| \\ = [\sum_{i=1}^n \sqrt{(\gamma_i^L p_i - \gamma_j^L p_j)^2}, \sum_{i=1}^n \sqrt{(\gamma_i^U p_i - \gamma_j^U p_j)^2}] \quad (21)$$

where $d(\tilde{h}_1(p), \tilde{h}_2(p))$ represents an interval value denoted as d and $i = j$.

3.2. Fuzzy integral and shapley value

Fuzzy measure and fuzzy integral are effective tools to measure the interaction among criteria. The λ -fuzzy measure and Shapley value are further proposed based on the fuzzy measure and fuzzy integral, which can simplify the fuzzy measure calculation and obtain the interaction criterion weights (Guo et al., 2020; Zhang et al., 2017). The concepts of fuzzy measure, λ -fuzzy measure and Shapley value are defined as follows.

Definition 11. (Wei et al., 2017) X is a measurable universe. A fuzzy measure μ defined on X is a function $\mu: F(X) \rightarrow [0,1]$ only if the following formulas are satisfied:

$$\mu(\emptyset) = 0, \mu(X) = 1; \\ \text{for all } A, B \in F, \text{ if } A \subseteq B \text{ then } \mu(A) \leq \mu(B). \quad (22)$$

Definition 12. (Guo et al., 2020) As a special kind of fuzzy measure, the additional condition needs to be satisfied for λ -fuzzy measure:

$$\mu(A \cup B) = \mu(A) + \mu(B) + \lambda \mu(A) \mu(B) \quad (23)$$

where for all $A, B \in F(X)$ and $A \cap B \in \emptyset$, $\lambda \in (-1, +\infty)$.

For a λ -fuzzy measure between criteria A and B . If $\lambda = 0$, then $\mu(A \cup B) = \mu(A) + \mu(B)$, μ become an additive measure and there exhibits no interaction between A and B . If $-1 < \lambda < 0$, then $\mu(A \cup B) < \mu(A) + \mu(B)$, μ become a sub-additive measure and there exhibits superfluous interaction between A and B . If $\lambda > 0$, then $\mu(A \cup B) > \mu(A) + \mu(B)$, μ become a super-additive measure and there exhibits mutualistic interaction between A and B .

For a finite set X , $\bigcup_{i=1}^n x_i = X$. Its λ -fuzzy measure can be computed by Eq. (24).

$$\mu(X) = \mu(\bigcup_{i=1}^n x_i) = \begin{cases} \frac{1}{\lambda} \left(\prod_{i=1}^n [1 + \lambda \mu(x_i)] - 1 \right) \lambda \neq 0. \\ \sum_{i=1}^n \mu(x_i) \lambda = 0. \end{cases} \quad (24)$$

where, for all $i, j = 1, 2, \dots, n$, $x_i \cap x_j = \emptyset$ and $x_i \neq x_j$. For a single element x_i , its λ -fuzzy measure $\mu(x_i)$ is called fuzzy density. $\mu(x_i)$ represents the membership degree of i th criterion, which is given by experts subjectively.

For any $A \in F(X)$, its λ -fuzzy measure can be computed by Eq. (25).

$$\mu(A) = \begin{cases} \frac{1}{\lambda} \left(\prod_{x_i \in A} [1 + \lambda \mu(x_i)] - 1 \right) \lambda \neq 0. \\ \sum_{x_i \in A} \mu(x_i) \lambda = 0. \end{cases} \quad (25)$$

For a special condition, if $A = X$, then $\mu(A) = \mu(X) = 1$. The value λ can be computed by Eq. (26).

$$\lambda + 1 = \prod_{i=1}^n [1 + \lambda \mu(x_i)] \quad (26)$$

where $\lambda \in (-1, +\infty)$ and $\lambda \neq 0$.

Definition 13. (Tan et al., 2015) The discrete generalized Choquet integral of real-valued function f to fuzzy measure λ on X is defined as follows:

$$C_\mu(f) = \sum_{i=1}^t f(x_i) [\mu(B_i - B_{i-1})] + \sum_{i=t+1}^n f(x_i) [\mu(A_i - A_{i+1})] \quad (27)$$

where i represents a permutation on X satisfying the condition $f(x_1) \leq \dots \leq f(x_t) < 0 \leq f(x_{t+1}) \leq \dots \leq f(x_n)$, and $B_i = \{x_1, \dots, x_i\}$, $B_0 = \emptyset$, $A_i = \{x_{i+1}, \dots, x_n\}$, $A_{n+1} = \emptyset$.

Shapley value, proposed on the basis of λ -fuzzy measure, is viewed as an effective tool to deal with the interactions in criteria, which also represents the overall importance of each criterion. Shapley value is defined by Eq. (28).

$$\mu_s(\mu, X) = \sum_{T \subseteq X \setminus A} \frac{(n-a-t)!t!}{(n-a+1)!} [\mu(A \cup T) - \mu(T)] \quad (28)$$

where X is the whole criterion set and $A \in F(X)$. T is a subset of $X \setminus A$, which represents the difference set of A and X . n , a and t represent the cardinal number of X , A , and T , respectively.

For $A = \{i\}$, where i represents any single set in X . The criterion weight of i th criterion $\mu_i(\mu, X)$ can be obtained using Eq. (29).

$$\mu_i(\mu, X) = \sum_{T \subseteq X \setminus i} \frac{(n-t-1)!t!}{n!} [\mu(A \cup T) - \mu(T)] \quad (29)$$

4. Multi-stage decision-making method

4.1. Shapley value method based on PIVHFS

The assessment information is collected in the form of the interval value hesitant fuzzy set by the fuzzy linguistic scale shown in Table 1.

Table 1

The fuzzy linguistic scale for expert assessment.

Linguistic term	Abbreviations	Fuzzy rating
Very low influence	VL	(0.0, 0.2)
Low influence	L	(0.2, 0.4)
Medium influence	M	(0.4, 0.6)
High influence	H	(0.6, 0.8)
Very high influence	VH	(0.8, 1.0)

The frequency of fuzzy linguistic scale used by experts is regarded as the probabilistic information of the interval value hesitant fuzzy evaluation. In this way, the assessment information from experts is transformed into the form of PIVHFS. For example, if the evaluation information for criterion P_1 given by four experts is $\{[0.6, 0.8]\}$, $\{[0.4, 0.6]\}$, $\{[0.6, 0.8]\}$, $\{[0.2, 0.4]\}$, then the evaluation information can be expressed in the form of PIVHFE as $\{[0.2, 0.4] (0.25), [0.4, 0.6] (0.25), [0.6, 0.8] (0.5)\}$. To settle criterion weights, the PIVHFS-based Shapley value is introduced, and the special steps are shown as follows.

Step 1. Determine the value of parameter λ . Collect the assessment information in the form of PIVHFS. By using Eq. (19), the fuzzy measures can be simplified as interval value numbers. According to Eqs. (23) and (26), the λ value can be calculated.

Step 2. Compute the fuzzy measure of the criterion combination. Based on Eq. (25), the fuzzy measure of criterion combination can be calculated.

Step 3. Obtain the Shapley weight of the criteria. According to the obtained fuzzy measures and Eq. (29), the PIVHFS-Shapley weight of criteria can be calculated.

4.2. TODIM-TOPSIS method based on PIVHFS

To address the material selection issue, a PIVHFS-based TODIM-TOPSIS approach is proposed. The decision matrix is constructed by the evaluation of experts and the experiment data in the form of PIVHFS. Next, the aggregation process and the normalization process are conducted to obtain the normalized decision matrix. Then, the TODIM method is applied to obtain the dominance degree of one alternative to another alternative under each criterion and further construct the dominance matrix under each criterion. By changing the aggregation process of the TODIM method, the overall dominance matrix, which contains the overall dominance of each criterion, can be obtained. In this way, the dominance between alternatives is transformed into the dominance between criteria, which includes the correlation information of criteria. Finally, the TOPSIS method is proposed to compute the distance measure of each alternative to the NIS and PIS, and further rank the alternatives.

Step 1. Construct the total decision matrix.

The decision information, which is processed in the way described in Section 3.1, is presented using decision matrices. The total decision matrix is constructed by adding up all the initial matrix using the score function of PIVHFEs. The total decision matrix is constructed of m alternatives and n criteria as shown in Eq. (30).

$$[D_{ij}]_{m \times n} = \begin{matrix} & \begin{matrix} C_1 & C_2 & \cdots & C_n \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} & \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{m1} & d_{m2} & \cdots & d_{mn} \end{bmatrix} \end{matrix} \quad (30)$$

Step 2. Normalize the total decision matrix.

Normalize the total decision matrix by transforming the cost criteria into benefit criteria by Eq. (31).

$$d_{ij} = \begin{cases} \frac{d_{\max_j} - d_{ij}}{d_{\max_j} - d_{\min_j}}, & \text{for cost - criteria} \\ \frac{d_{ij} - d_{\min_j}}{d_{\max_j} - d_{\min_j}}, & \text{for benefit - criteria} \end{cases} \quad (31)$$

where $d_{\max_j}$ and $d_{\min_j}$ are two vectors, which represent the maximum and minimum value of d_{ij} , respectively.

Step 3. Construct the dominance matrix

For criterion C_k , the relative dominance degree of one alternative A_i on another alternative A_j can be computed by Eq. (32).

$$\phi_c(A_i, A_j) = \begin{cases} \sqrt{\frac{w_{kr}}{\sum_{k=1}^n w_{kr}}} (d_{ik} - d_{jk}), & \text{if } (d_{ik} - d_{jk}) > 0 \\ 0, & \text{if } (d_{ik} - d_{jk}) = 0 \\ -\frac{1}{\theta} \sqrt{\frac{\sum_{k=1}^n w_{kr}}{w_{kr}}} (d_{jk} - d_{ik}), & \text{if } (d_{ik} - d_{jk}) < 0 \end{cases} \quad (32)$$

where $w_{kr} = w_k/w_r$, which represents the relative weight of criterion C_k to criterion C_r . w_r is a reference weight vector, which can be calculated by $w_r = \max(w_j)$. θ denotes a recession coefficient.

By computing the relative dominance degree of alternatives, the dominance matrix under each criterion can be built. There are n dominance matrices and the dominance matrix under criterion C_k can be presented by Eq. (33).

$$[\phi_k(A_i, A_j)] = \begin{bmatrix} 0 & \phi_k(A_1, A_2) & \cdots & \phi_k(A_1, A_m) \\ \phi_k(A_2, A_1) & 0 & \cdots & \phi_k(A_2, A_m) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_k(A_m, A_1) & \phi_k(A_m, A_2) & \cdots & 0 \end{bmatrix} \quad (33)$$

Step 4. Construct the overall dominance matrix.

By adding up all relative dominance, the overall dominance degree can be constructed. For criterion C_k and alternative A_i , the overall dominance degree is defined by Eq. (34).

$$\Phi_k(A_i) = \sum_{j=1}^m [\phi_k(A_i, A_j)]_{m \times 1} \quad (34)$$

By computing the overall dominance degree, the dominance matrices are changed into an overall dominance vector under criterion C_k . Therefore, for m alternatives, there are n dominance matrices become overall dominance vectors, which come up with a new decision matrix denoted as the overall dominance matrix. In this way, the correlation between criteria can be characterized by the dominance degree of criteria. The overall dominance matrix can be obtained by Eq. (35).

$$[\Phi_j(A_i)]_{m \times n} = \begin{bmatrix} 0 & \Phi_2(A_1) & \cdots & \Phi_n(A_1) \\ \Phi_1(A_2) & 0 & \cdots & \Phi_n(A_2) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_1(A_m) & \Phi_2(A_m) & \cdots & 0 \end{bmatrix} \quad (35)$$

Step 5. Construct the weighted normalized decision matrix.

Normalize the overall dominance matrix based on the normalization rules of TOPSIS method, as shown in Eq. (36). The weighted normalized decision matrix is defined by Eq. (37).

$$r_{ij} = \Phi_j(A_i) / \sqrt{\sum_{i=1}^m \Phi_j(A_i)^2}, [r_{ij}]_{m \times n} \quad (36)$$

$$v_{ij} = w_j * r_{ij}, [v_{ij}]_{m \times n} \quad (37)$$

Step 6. Determine the distance measure of each alternative.

The positive distance measure can be defined by adding up the distance of each element to the PIS. Similarly, the negative distance measure can be defined by adding up the distance of each element to the NIS. PIS is a vector containing the maximum value under each criterion in the overall decision matrix and NIS is the opposite. The ideal solution (C_j^+ and C_j^-) and the distance measure (D_i^+ and D_i^-) can be calculated by Eqs. (38) and (39).

$$\begin{aligned} C_j^+ &= \max([v_{ij}]_{m \times n}) \\ C_j^- &= \min([v_{ij}]_{m \times n}) \end{aligned} \quad (38)$$

$$\begin{aligned} D_i^+ &= \sqrt{\sum_{j=1}^n (v_{ij} - C_j^+)^2}, i = 1, 2, \dots, m \\ D_i^- &= \sqrt{\sum_{j=1}^n (v_{ij} - C_j^-)^2}, i = 1, 2, \dots, m \end{aligned} \quad (39)$$

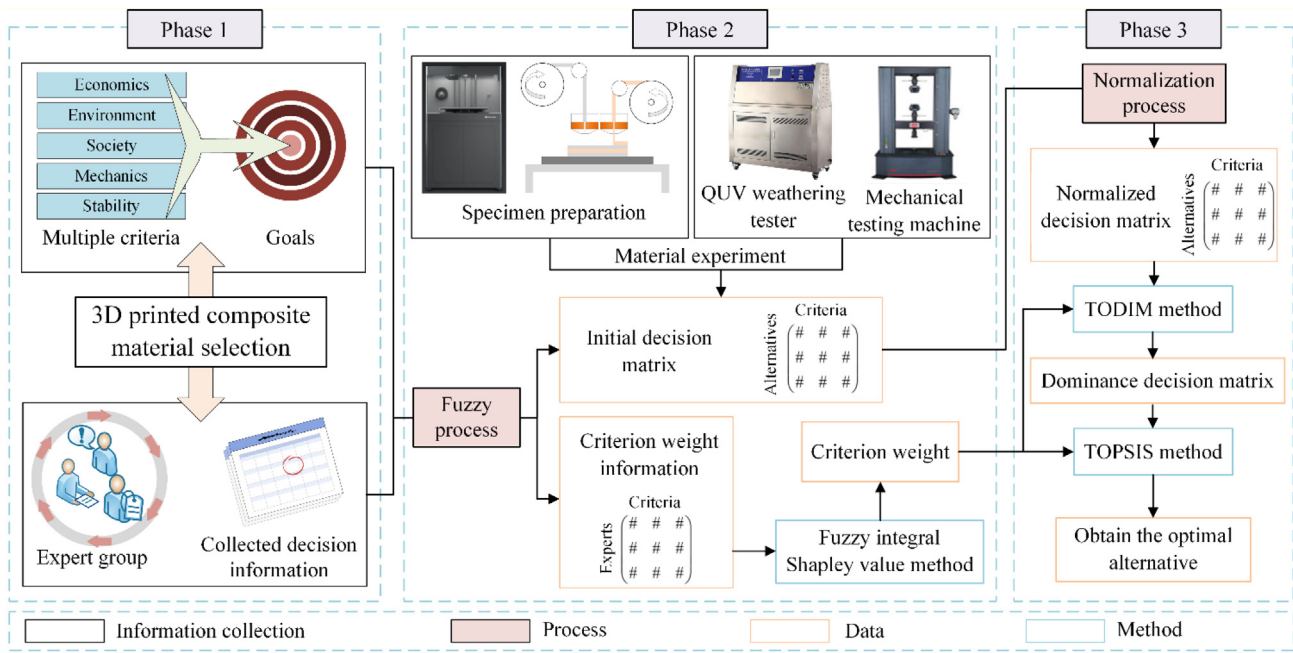


Fig. 1. Framework of the proposed decision-making method.

Step 7. Determine the relative proximity.

The relative proximity of each alternative can be obtained by Eq. (40).

$$R_i = \frac{D_i^-}{D_i^- + D_i^+} \quad (40)$$

where the larger value of R_i represents the better alternative.

Step 8. Rank the alternatives and select the optimal alternative.

4.3. The frame of the proposed method

Based on the abovementioned methods, the research frame of the proposed multi-stage decision-making method with PIVHFS for 3D printed composite material selection is presented. The framework consists of three main phases and the specific content of each phase is shown as follows

Phase 1. Establish the hierarchical multiple criterion model and collect decision information.

A hierarchical multiple criterion model considering economic, environmental, social, mechanical, and probabilistic information is established for 3D printed composite material selection. Numerous qualified stakeholders and experts in different positions are invited to offer the decision information, which includes the decision information of the decision matrix and criterion weight.

Phase 2. Obtain the criterion weight and initial decision matrix.

The assessment information obtained from Phase 1 is fuzzed and converted into the form of PIVHFS to quantize the expert opinion and characterize the hesitant, uncertain, and probabilistic information. The fuzzy integral integral-based method, that is Shapley value, considering criterion interaction effects is proposed to compute the criterion weight. In addition, the mechanical experiments of 3D printed composite material in hygrothermal aging environments are conducted to obtain the mechanical criteria. The initial decision matrix including assessment information and mechanical information is established.

Phase 3. Select the best alternative.

By normalizing the initial decision matrix, the normalized decision matrix with m alternative and n criteria is obtained. In the decision process, the criterion weights are used twice, in the TODIM method and TOPSIS method respectively. Based on the normalized decision matrix,

the TODIM method is introduced to characterize the alternative dominance and further establish the dominance decision matrix, in which the criterion weights are used to represent weights of the dominance degree between different alternatives. In addition, the TOPSIS method incorporating the criterion weights is applied to address the optimal alternative.

The framework of the proposed decision-making method is shown in Fig. 1.

5. Case study

5.1. Engineering background and material preparation

5.1.1. Engineering background

In recent years, composite material is engaged used in automotive components for its lightweight and high strength (Behera et al., 2023; Mohammadi et al., 2023). The composites-made automotive component can effectively improve fuel efficiency, and reduce carbon emissions, which has drawn a lot of attention (Juraeva et al., 2017; Vaidya et al., 2023). The automotive chassis, as an important component of the automobile, is responsible for loading-bearing, component support, and protection of automobile parts (Betancur et al., 2017; Shah et al., 2023). There have been a series of studies focusing on the application of composite materials in automotive chassis (Coppola et al., 2023; Ishikawa et al., 2018). With the development of advanced manufacturing, the application of 3D printed composites in automotive components is highly anticipated (Huang et al., 2023). Therefore, the automotive chassis is taken as an engineering background in this study to settle the 3D printed composite material selection problem. In this study, three kinds of 3D printed composites with different printing parameters and materials were prepared. A three-point bending test was conducted on the composites combined with the loading-bearing characteristics of the automobile chassis. In addition, a hygrothermal aging test was conducted on the composites to simulate the external environments such as high temperatures and rainy weather of the automobiles.

5.1.2. Material preparation

To enhance the overall mechanical properties of composites, the introduction of short and continuous fibers in polymer composites

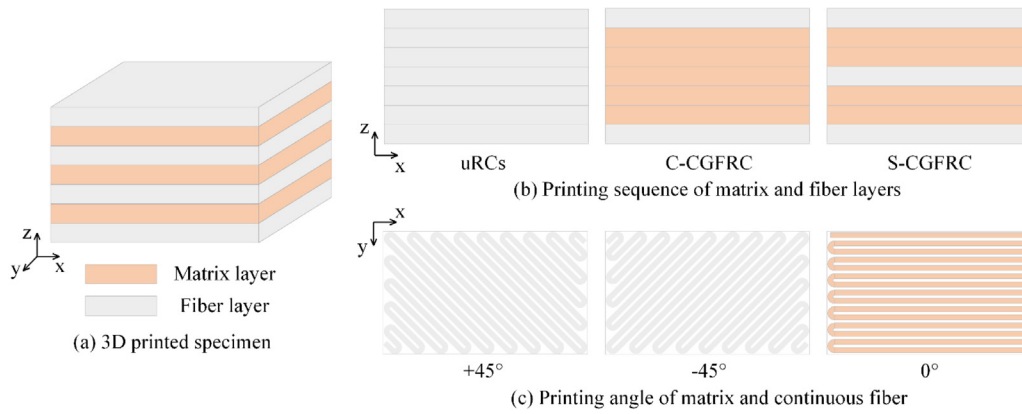


Fig. 2. (a) specimen of 3D printed continuous glass fiber composites, (b) printing sequence of matrix and continuous fiber, (c) printing angle of matrix and continuous fiber.

Table 2

Original evaluation information for criteria.

Criteria	Grading	Criteria	Grading	Criteria	Grading
P ₁	2VH, 6H	F ₆	3M, 5L	F ₁₆	2VH, 4H, 2M
P ₂	7M, 1L	F ₇	3H, 4M, 1L	F ₁₇	3VH, 5H
P ₃	5H, 3M	F ₈	3H, 4M, 1VL	F ₁₈	2VH, 4H, 2M
P ₄	8VH	F ₉	5VH, 3H	N ₁	8H
P ₅	4VH, 4H	F ₁₀	1VH, 7H	N ₂	8H
F ₁	7VH, 1H	F ₁₁	8M	N ₃	8H
F ₂	1VH, 2H, 3M, 2L	F ₁₂	1VH, 6H, 1L	N ₄	8H
F ₃	1H, 3M, 3L, 1VL	F ₁₃	4VH, 2H, 2M	N ₅	6H, 2M
F ₄	1H, 5M, 2L	F ₁₄	5VH, 3H	N ₆	6H, 2M
F ₅	1VH, 6H, 1M	F ₁₅	3VH, 4H, 1M		

has been viewed as an effective way (Li et al., 2023; Zhu et al., 2023). In this study, printing filaments are produced by Markforged® (Cambridge, MA, USA). Short carbon fiber reinforced polyamide with a content of 15.00 wt. % short carbon fibers is chosen as an effective matrix. Continuous glass fiber impregnated with polyamide, and the content of glass fiber was 50.30 wt. %, is chosen as reinforcement. Specimens are manufactured by a Markforged® X7 (Cambridge, MA, USA) 3D printer on the basis of fused deposition modeling (FDM) principles. During the manufacturing process, continuous fiber filaments and effective matrix filaments are heated to a molten state and deposited on the printing platform by fiber and matrix nozzle, respectively. The deposited filaments are settled line by line to form a plane and then plane by plane to build the final designed specimens reinforced by continuous fiber layers (SCGFL).

In this study, the specimens are designed 4.6 mm in height, 6 mm in width, and 90 mm in length as shown in Fig. 2(a). Three types of specimens are manufactured as shown in Fig. 2(b), that is specimens without continuous fiber reinforced (uRCs), concentrated stacking specimens reinforced by continuous glass fiber layers (CS-SCGFL), and separated stacking specimens reinforced by continuous glass fiber layers (SS-SCGFL). Continuous glass fibers and effective matrix layers are printed with 0° and ±45° raster angles in the horizontal plane, respectively, as shown in Fig. 2(c). Each specimen has 46 layers with a single-layer thickness of 0.1 mm. Both CS-SCGFL and SS-SCGFL have 30 layers of reinforced fiber.

5.2. The weight vector of each criterion

5.2.1. Multiple criteria model

A hierarchical multiple criterion model of 3D printed composites is established as shown in Fig. 3. The target level (E) is the material selection of 3D printed composites in the hygrothermal aging environments. The alternative level consists of three types of specimens presented in Section 3.1.2. Three criterion levels (P, F, N) are established, which include economic criterion (P₁), environment criterion (P₂), society criterion (P₃), mechanical criterion (P₄), and stability

criterion (P₅) in the first level. The second criterion level contains the subdivision criteria of the first level, which includes material cost (F₁), manufacturing cost (F₂), maintenance cost (F₃), recovery cost (F₄), environmental protection (F₅), raw materials extraction (F₆), recycling potential (F₇), air pollution (F₈), light-weight level (F₉), operational life (F₁₀), production efficiency (F₁₁), health and safety (F₁₂), flexural modulus (F₁₃), flexural strength (F₁₄), flexural energy absorption (F₁₅), and the stability criterion of flexural modulus (F₁₆), flexural strength (F₁₇), and flexural energy absorption (F₁₈). In addition, the third criterion level (N₁–N₆) represents the stability of flexural modulus, flexural strength, and flexural energy absorption in 15 and 30 days. The criteria above can be divided into two categories: the benefit criterion and the cost criterion. For the benefit criterion, the large value is desirable, which is opposite for the cost criterion. In the multiple criterion model, the benefit criterion includes F₅, F₇, F₁₀–F₁₅, and N₁–N₆, and the cost criterion includes F₁–F₄, F₆, F₈, and F₉.

5.2.2. Obtain the weight vector of each criterion

For diverse printing strategies, the weight vectors of conflicting criteria need to be settled. To implement the material selection of 3D printed composites, eight qualified stakeholders and experts (E₁ – E₈) in different positions are invited to give their opinions on three layers of criterion (P layer, F layer, and N layer). The original evaluation information is collected as shown in Table 2. The probabilistic interval value hesitant fuzzy information can be further obtained as shown in Table A1. According to the types of criteria, the criteria, which belong to the same layer or the same category, are divided into several groups. Then, the PIVHFS-based Shapley method is introduced to compute the criterion weight of each group. Namely, the groups include P₁–P₅, F₁–F₄, F₅–F₈, F₉–F₁₂, F₁₃–F₁₅, F₁₆–F₁₈, N₁–N₂, N₃–N₄, and N₅–N₆. Take group F₁–F₅ as an example, its criterion weight is calculated as follows.

Step 1. For criteria F₁–F₅, its fuzzy measure is given by eight experts as: {[0.8, 1](0.875), [0.6, 0.8](0.125)}, {[0.8, 1](0.125), [0.6, 0.8](0.25)}, [0.4, 0.6](0.375), [0.2, 0.4](0.25)}, {[0.6, 0.8](0.125), [0.4, 0.6](0.375)}, [0.2, 0.4](0.375), [0, 0.2](0.125)}, {[0.6, 0.8](0.125), [0.4, 0.6](0.625)}, [0.2, 0.4](0.25)}. According to Eq. (19), the fuzzy

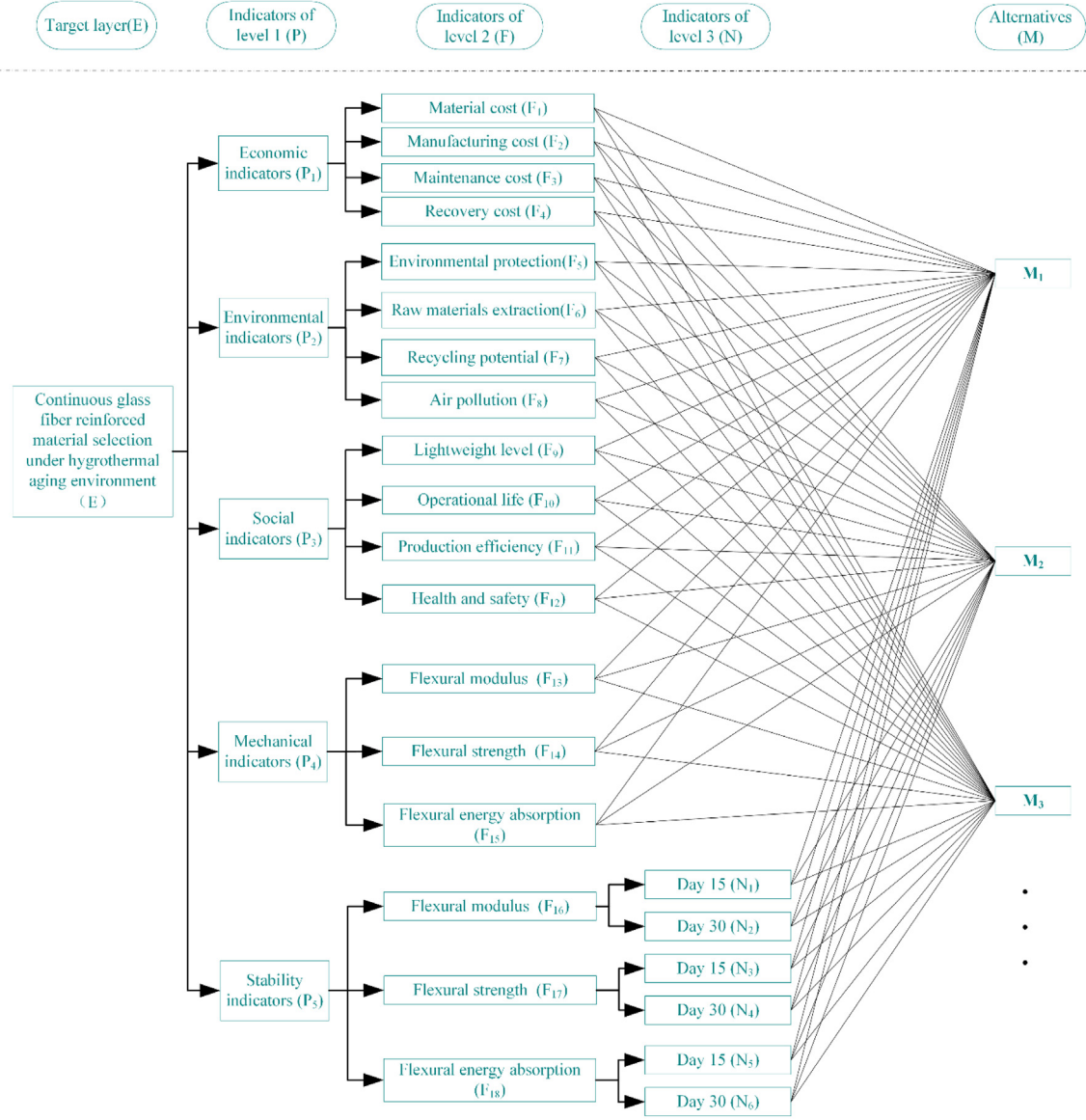


Fig. 3. Hierarchical multiple criteria model.

measures can be simplified as interval value numbers, which are $[0.775, 0.975]$, $[0.45, 0.65]$, $[0.3, 0.5]$, $[0.375, 575]$. According to Eqs. (23) and (26), the λ value calculated is: $\lambda = [-99798, -0.92038]$.

Step 2. Let T be the criteria combination, and $\mu(T)$ be its fuzzy measure. According to Eq. (25), the fuzzy measure $\mu(T)$ can be calculated, as shown in Table 3.

Step 3. According to the obtained fuzzy measures and Eq. (29), the PIVHFS-Shapley weight of criteria can be calculated as follows: $\mu(F_1) = [0.4097, 0.4572]$, $\mu(F_2) = [0.2229, 0.2281]$, $\mu(F_3) = [0.1399, 0.1661]$, $\mu(F_4) = [0.1800, 0.1961]$.

In this way, the criterion weight of other groups can also be obtained. The criterion weights of each group are obtained as shown in Table 4. The final criterion weight can be obtained by weighting the criterion weight of each layer in the hierarchical multiple criterion model, the results are shown in Fig. 4.

It can be found that material cost (F₁), light-weight level (F₉), and mechanical properties (F₁₃–F₁₅) exhibit high importance, while the maintenance cost (F₃), raw material extraction (F₆), and production efficiency (F₁₁) exhibit low importance. In general, mechanical properties are the most important, while environmental criteria have the least important. The criteria in stability aspects exhibit relative

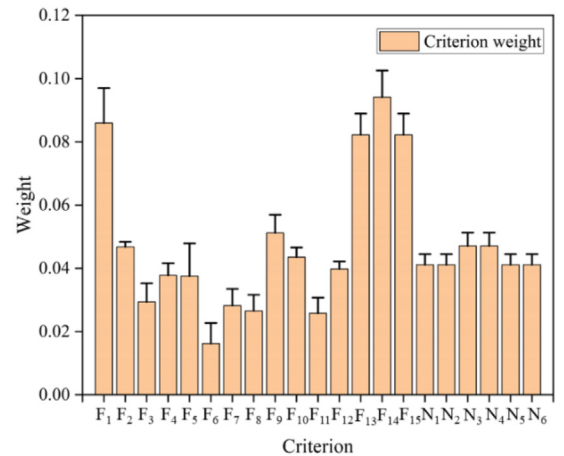


Fig. 4. The weight of each criterion.

Table 3
The fuzzy measure of criteria combination.

Criteria	Fuzzy measure	Criteria	Fuzzy measure	Criteria	Fuzzy measure
F ₁	[0.775, 0.975]	F ₁ , F ₃	[0.8610, 0.9885]	F ₁ , F ₂ , F ₃	[0.9544, 0.9973]
F ₂	[0.45, 0.65]	F ₁ , F ₄	[0.8825, 0.9905]	F ₁ , F ₂ , F ₄	[0.9670, 0.9980]
F ₃	[0.3, 0.5]	F ₂ , F ₃	[0.6257, 0.8257]	F ₁ , F ₃ , F ₄	[0.9388, 0.9963]
F ₄	[0.375, 0.575]	F ₂ , F ₄	[0.6697, 0.8520]	F ₂ , F ₃ , F ₄	[0.7848, 0.9269]
F ₁ , F ₂	[0.9040, 0.9925]	F ₃ , F ₄	[0.5715, 0.7881]	F ₁ , F ₂ , F ₃ , F ₄	[1, 1]

Table 4
The PIVHFS-Shapley weight of criteria.

Criteria	Weight	Criteria	Weight	Criteria	Weight
P ₁	[0.2098, 0.2122]	F ₆	[0.1448, 0.1719]	F ₁₆	[0.3153, 0.3183]
P ₂	[0.1118, 0.1318]	F ₇	[0.2524, 0.2541]	F ₁₇	[0.3634, 0.3694]
P ₃	[0.1640, 0.1726]	F ₈	[0.2375, 0.2401]	F ₁₈	[0.3153, 0.3183]
P ₄	[0.2601, 0.2788]	F ₉	[0.3123, 0.3300]	N ₁	[0.1700, 0.1723]
P ₅	[0.2257, 0.2332]	F ₁₀	[0.1719, 0.2700]	N ₂	[0.1700, 0.1723]
F ₁	[0.4097, 0.4572]	F ₁₁	[0.1573, 0.2524]	N ₃	[0.1700, 0.1723]
F ₂	[0.2229, 0.2281]	F ₁₂	[0.2401, 0.2427]	N ₄	[0.1700, 0.1723]
F ₃	[0.1399, 0.1661]	F ₁₃	[0.3161, 0.3190]	N ₅	[0.1554, 0.1600]
F ₄	[0.1800, 0.1961]	F ₁₄	[0.3620, 0.3678]	N ₆	[0.1554, 0.1600]
F ₅	[0.3356, 0.3636]	F ₁₅	[0.3161, 0.3190]		

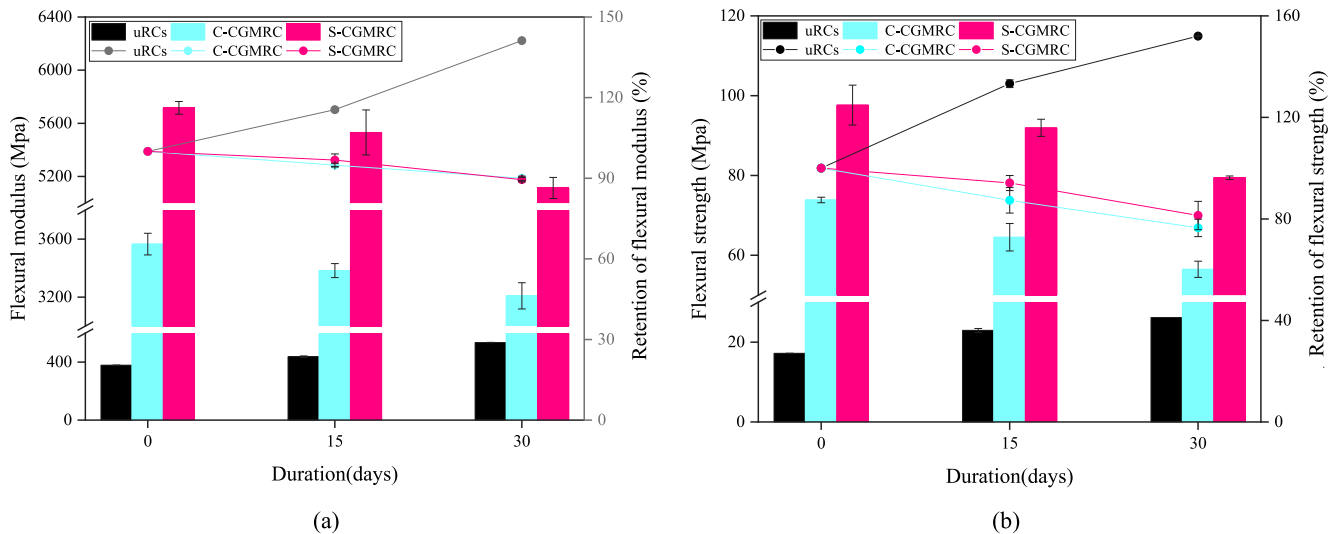


Fig. 5. (a) Flexural modulus and retention of the specimens; (b) Flexural strength and retention of the specimens.

equilibrium importance, and its overall importance level is second only to the mechanical properties.

5.3. Evaluation of the material alternatives

5.3.1. Establish the decision matrix

By carrying out mechanical investigation and collecting expert assessment information, the decision matrix can be constructed. Based on the material preparation, the mechanical experiment in hygrothermal aging environments can be carried out. A hygrothermal aging investigation is conducted to simulate the natural aging process using a weathering tester (Q-Lab, Westlake, OH, USA) with the condition of 60 °C and 100% relative humidity. A variety of timeframes, up to 30 days, are spent testing the samples. A universal mechanical testing machine (E44, MTS Co., USA) is used to perform three-point bending testing and further investigate the mechanical properties of as-printed and aged specimens. The indenter speed of three-point bending testing is 5 mm/min. As the test progresses, stress-strain curves are recorded, from which the flexural modulus and flexural strength can be obtained. The final results are the average of three repeated tests. The flexural modulus (F₁₃), flexural strength (F₁₄), flexural energy absorption (F₁₅), and corresponding retention (N₁–N₆) of uRCs, CS-SCGFL, and SS-SCGFL at various hygrothermal aging durations are shown in Fig. 5.

The results demonstrate that the flexural modulus and strength of CS-SCGFL and SS-SCGFL are greatly enhanced by continuous glass fiber compared with uRCs. For as-prepared and aged SCGFL, SS-SCGFL has a higher flexural modulus and strength than CS-SCGFL. For uRCs, the flexural modulus and strength increase with longer aging times during the aging phase, which lasts from 0 to 30 days. Its modulus and strength are retained at a higher level after 15 and 30 days of aging, reaching 141.23% and 152.05%, respectively. While for CS-SCGFL and SS-SCGFL, their modulus and strength deteriorate with aging time. The modulus and strength retention for CS-SCGFL drop to 90.01% and 76.49%, respectively, whereas they drop to 89.46% and 81.36% for SS-SCGFL. The conclusion can be drawn that the specimens with separated continuous glass fiber layers had the highest flexural strength, however, their aging resistance had no significant advantage.

Except for the mechanical criteria, some criteria, e.g., the material cost, manufacturing cost, and production efficiency of the target printed specimen can be estimated by the slicing software Eiger[®]. In addition, the mass of each specimen can be obtained by an electronic scale. The results are summarized in Table 5. The material cost and mass of SCGFLs are higher than that of Matrix, and the printing time of SCGFLs was longer. The rationale is that continuous glass fiber is slower printing and more expensive than Onyx[®], as well as being heavier. In addition, for SCGFLs, the layer stacking sequence has little effect on

Table 5

Material cost, printing time, and mass of specimens.

Linguistic term	uRCs	CS-SCGFLs	SS-SCGFLs
Material cost	0.62	2.2	2.21
Printing time	2040	2880	2880
Production efficiency	1.765	1.25	1.25
Weight	2.8033	2.88	2.8833

their initial costs and printing efficiencies. The assessment information of the rest criteria is evaluated by experts in the form of PIVHFS, which include F_3 – F_8 , F_{10} , and F_{12} . By the abovementioned ways, the decision matrix is finally established. The assessment information in the decision matrix is expressed as PIVHFS as shown in Table A2.

5.3.2. Decision making based on TODIM-TOPSIS method

The initial decision matrix is established in Section 5.3.1. The total decision matrix is obtained by transforming the PIVHFS into interval value numbers. Normalize the total decision matrix by transforming the cost criteria into benefit criteria. After normalization, the interval value is limited to 0 to 1.

The correlation weight of each criterion can be computed as $w_{1-14} = [0.9129, 0.9461]$, $w_{2-14} = [0.4720, 0.4967]$, $w_{3-14} = [0.3117, 0.3437]$, $w_{4-14} = [0.4011, 0.4058]$, $w_{5-14} = [0.3985, 0.4673]$, $w_{6-14} = [0.1719, 0.2209]$, $w_{7-14} = [0.2997, 0.3266]$, $w_{8-14} = [0.2820, 0.3086]$, $w_{9-14} = [0.5440, 0.5555]$, $w_{10-14} = [0.4545, 0.4624]$, $w_{11-14} = [0.2740, 0.2994]$, $w_{12-14} = [0.4112, 0.4227]$, $w_{13-14} = [0.8673, 0.8732]$, $w_{14-14} = [1.000, 1.000]$, $w_{15-14} = [0.8673, 0.8732]$, $w_{N1-14} = [0.4337, 0.4366]$, $w_{N2-14} = [0.4337, 0.4366]$, $w_{N3-14} = [0.5000, 0.5000]$, $w_{N4-14} = [0.5000, 0.5000]$, $w_{N5-14} = [0.4337, 0.4366]$, $w_{N6-14} = [0.4337, 0.4366]$.

In addition, with the recession coefficient $\theta = 1$, the dominance degree for each criterion can be computed. For brevity, an example of the dominance degree matrices of criterion F_1 – F_4 is shown as follows:

$$\phi_{F_1} = \begin{pmatrix} [0, 0] & [0.2943, 0.2967] & [0.2953, 0.2976] \\ [-3.3760, -3.3494] & [0, 0] & [0.0234, 0.0236] \\ [-3.3867, -3.3599] & [-0.2686, -0.2665] & [0, 0] \end{pmatrix},$$

$$\phi_{F_2} = \begin{pmatrix} [0, 0] & [0.2102, 0.2178] & [0.2102, 0.2178] \\ [-4.7567, -4.5914] & [0, 0] & [0, 0] \\ [-4.7567, -4.5914] & [0, 0] & [0, 0] \end{pmatrix},$$

$$\phi_{F_3} = \begin{pmatrix} [0, 0] & [0.1543, 0.1605] & [0.1725, 0.1794] \\ [-5.1837, -4.9858] & [0, 0] & [0.0772, 0.0802] \\ [-5.7956, -5.5744] & [-2.5919, -2.4930] & [0, 0] \end{pmatrix},$$

$$\phi_{F_4} = \begin{pmatrix} [0, 0] & [0.1949, 0.1957] & [0.1887, 0.1895] \\ [-5.1304, -5.1094] & [0, 0] & [-1.2826, -1.2773] \\ [-4.9674, -4.9471] & [0.0487, 0.0489] & [0, 0] \end{pmatrix}.$$

The overall dominance matrix can be obtained by Eqs. (34) and (35) as illustrated in Table A3. The weighted normalized decision matrix can be computed using Eqs. (36) and (37). The PIS and NIS are obtained as shown in Table A4. The distance measure of each alternative can be computed as $D_1^+ = [0.1684, 0.1822]$, $D_2^+ = [0.2219, 0.2436]$, $D_3^+ = [0.2559, 0.2822]$, $D_1^- = [0.1436, 0.1607]$, $D_2^- = [0.1870, 0.2054]$, $D_3^- = [0.2526, 0.2757]$.

The relative proximity can be computed using Eq. (40), i.e., $R_1 = [0.4602, 0.4687]$, $R_2 = [0.4573, 0.4575]$, and $R_3 = [0.4941, 0.4967]$.

By comparing the relative proximity of each alternative, the final ranks can be drawn as $A_3 > A_1 > A_2$.

The results show that the SS-SCGFLs is the optimal alternative, and the CS-SCGFLs is the worst. By further analysis, it can be found that the addition of continuous glass fiber leads to higher mechanical properties, but also higher costs and lower environmental affinity. For specimens without continuous fiber reinforced, though its mechanical properties

are relatively low, the advantages in economic, environmental and stability aspects give it an edge in the decision-making process.

Compared with CS-SCGFLs and uRCs, SS-SCGFLs has great advantages in mechanical properties and stability, making SS-SCGFLs the optimal alternative. Therefore, it can be found that the addition of continuous fibers may not improve the dominance of 3D printed composites. However, by designing different stacking strategies, the continuous fiber-reinforced 3D printed composites can obtain a high advantage in the process.

6. Discussion

6.1. Comparative analysis

6.1.1. Comparative analysis of fuzzy TODIM-TOPSIS method and fuzzy TODIM/TOPSIS method

Two common decision methods, that are PIVHFS-based TODIM method (T_1) and the PIVHFS-based TOPSIS method (T_2), are proposed to make the comparative analysis of the PIVHFS-based TODIM-TOPSIS method (T_3) and further illustrate the scientificity and feasibility of the proposed approach. The analysis is carried out using identical criteria weights and engineering backgrounds. Table 6 represents the ranking results.

Based on the result of Table 6, two conclusions can be conducted. (1) The optimal alternative of T_1 and T_3 is A_3 . The difference is that A_2 in T_1 ranks higher than that in T_3 , while A_1 is on the contrary. Compared with T_3 , T_1 only considers the dominance degree between two alternatives and neglects the distance measure of all alternatives at the global scale. Therefore, the huge dominance gap between A_1 and A_2 in economic, environmental, and stability aspects is not fully considered, making A_1 lower in T_1 than that in T_3 . (2) The ranking results of T_2 and T_3 are completely different. In T_2 , A_1 ranks higher than A_2 and A_3 , and A_2 exhibits a similar level to A_3 . For A_1 has great strength in economic, environmental and stability aspects, the obtained distance measure of A_1 presents obvious dominance. However, the distance measure can only simply reflect the strengths of the alternatives, neglecting the relative dominance degree between alternatives. Therefore, T_2 cannot truly reflect the strength of alternatives. A conclusion can be drawn that the PIVHFS-based TODIM-TOPSIS method can consider both the global distance measure and relative dominance degree between alternatives.

6.1.2. Comparative analysis of the proposed method and ahp/fuzzy ahp method

The analytic hierarchy process (AHP) method is a typical MCDM method organized with hierarchical structures. In this section, the AHP method (M_1) and PIVHFS-based AHP method (M_2) are introduced to conduct a comparative analysis to illustrate their difference with PIVHFS-based Shapley value (M_3) and PIVHFS-based TODIM-TOPSIS (M_4). The comparative analysis is carried out under the consistent backgrounds of this study. Both the AHP method and fuzzy AHP pass the consistency check in the calculation process. The criterion weights of the three methods are computed as shown in Fig. 6. The criterion weights show that most criterion weights in M_3 are higher than that in M_1 and M_2 . The criterion weights in M_1 and M_2 are larger in high-value weights and smaller in low-value weights than in M_3 . It illustrates that the criterion weights in M_3 are more stable than that in M_1 and M_2 . While the criterion weights in M_1 and M_2 present large deviations and extreme values, which may result in inaccurate decision results. In the M_3 method, the interactions among criteria have been considered, which means each criterion weight has an effect on other criteria and also receives effects from other criteria, to make the weights more stable and thus avoid the extreme weight values. Therefore, though M_1 , M_2 , and M_3 are all organized with hierarchical structures, M_3 presents better advantages in the process of obtaining criterion weights.

The decision results of the M_1 , M_2 , and M_4 methods are shown in Table 7. It can be found that all methods rank A_3 as the optimal

Table 6
Ranking results of T_1 , T_2 , and T_3 .

Method	A_1		A_2		A_3	
	Results	Ranking	Results	Ranking	Results	Ranking
T_1	[0.4663, 0.4696]	3	[0.5064, 0.5095]	2	[1.0288, 1.0292]	1
T_2	[0.5584, 0.5635]	1	[0.4706, 0.4714]	2	[0.4633, 0.4655]	3
T_3	[0.4602, 0.4687]	2	[0.4573, 0.4575]	3	[0.4941, 0.4967]	1

Table 7
Ranking results of M_1 , M_2 and M_4 .

Method	A_1		A_2		A_3	
	Results	Ranking	Results	Ranking	Results	Ranking
M_1	0.3244	2	0.2468	3	0.4289	1
M_2	[0.2036, 0.3424]	3	[0.2669, 0.4438]	2	[0.2946, 0.4832]	1
M_4	[0.4602, 0.4687]	2	[0.4573, 0.4575]	3	[0.4941, 0.4967]	1

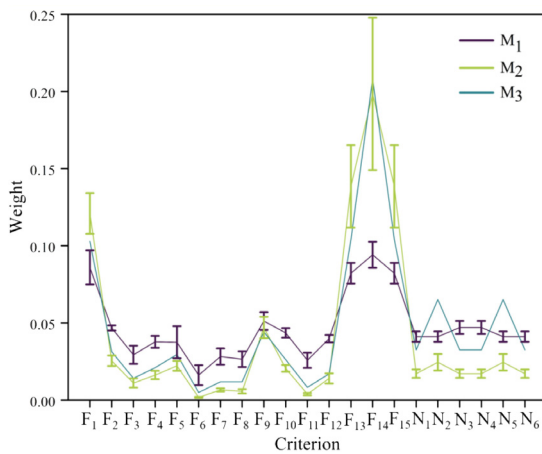


Fig. 6. Criterion weights of M_1 , M_2 , and M_3 .

alternative, but the worst method in M_2 is different from M_1 and M_4 . The reason is that mechanical properties in M_2 present larger weights while the economic, environmental, and social aspects present lower weights than that in M_4 . As A_1 dominates economic, environmental, and social aspects and A_2 emerges dominance in mechanical properties, the final ranking of A_2 is higher than A_1 in M_2 . As the criterion weights in M_1 and M_2 are slightly closer, the differences in their decision results illustrate that PIVHFS may greatly influence the pairwise comparison matrix of alternatives in the AHP method and further influence the decision results. In addition, the decision results in M_4 emerge more accurate interval values than that in M_2 . The above analysis illustrates the AHP method is prone to extreme values in solving the issues in this paper. In addition, the addition of PIVHFS in the AHP method will result in inaccurate decision results. Compared with AHP and fuzzy AHP method, the proposed approach in this study presents a more stable and accurate characteristic in the decision process.

6.2. Sensitivity analysis

To investigate the stability of the decision results, the sensitivity analysis of criterion weight w_j and recession coefficient θ is applied. The recession coefficient is used to express the sensitivity to the loss for decision-makers, the higher the θ is, the lower the loss avoidance. To test the effect of loss avoidance behavior on the final results, various levels of loss avoidance are set, which are 0.01, 0.05, 0.1, 0.5, 1, 2, 3, 4 and 5, the results are shown in Fig. 7(a). It can be found that with the increase of θ , A_2 , and A_3 decrease slowly, while A_1 increases. The changing trend indicates that lower loss avoidance tends to select the composites without the addition of continuous fibers. However, the

final ranking results do not vary with θ changes, which illustrates that the decision results are stable.

To explore the effects of criterion weight on decision results, the fluctuation analysis is carried for criterion weight. Comparatively to the initial criterion weight, each criterion has a 50% drop in weight. For each processed criterion, the decision results are respectively computed as shown in Fig. 7(b). It can be found that the ranking results are unchanged under criteria F_2 – F_{13} and N_1 – N_6 , but change under criteria F_1 , F_{14} , and F_{15} . For F_1 , A_1 has dominance in material cost, therefore, the weight reduction in F_1 will result in the ranking dropping for A_1 . A_3 has great dominance in mechanical properties (F_{13} – F_{15}). The weight reduction in mechanical properties makes the decision results of A_3 drop and A_1 improved, accordingly. In general, the fluctuation analysis of criterion weight illustrates that the decision results exhibit relatively stable character.

6.3. Managerial implications

Some valuable managerial implications can be provided in this section. This study has worked on the material selection issue of 3D printed composites considering the stability, mechanics, environment, economy, and society aspects. The proposed multi-stage decision-making method can help managers to select the optimal material and feedback to guide the manufacturing and design process of 3D printed composite materials. From the perspective of engineering application, the mechanical properties of freshly manufactured 3D printed composites and that of 3D printed composites in hygrothermal aging environments are both considered in this study, which is of more practical significance for the application of 3D printed composites in engineering fields. From an economic standpoint, the key economic indicators of 3D printed composites have been introduced to guide the design of more economical and high-strength composite materials. For the aspects of environment and society, the impact on the natural environment and human society during the production of composite materials is further considered. The research can be devoted to reducing the destruction of the natural environment and improving human social benefits. In addition, the PIVHFS proposed in this study can effectively address the fuzzy issues in material selection, e.g., the hesitancy of information, incomplete divergence, and inaccurate judgment. The proposed PIVHFS-based Shapley value and PIVHFS-based TODIM-TOPSIS approach can depict the interactions and correlation effects among criteria, which better meet the actual decision situation. This study provides a meaningful approach to solve material selection issues and a valuable reference for related research.

6.4. Research limitation

Material selection is a common issue in manufacturing. Though this study presents an effective MCDM approach to solving the

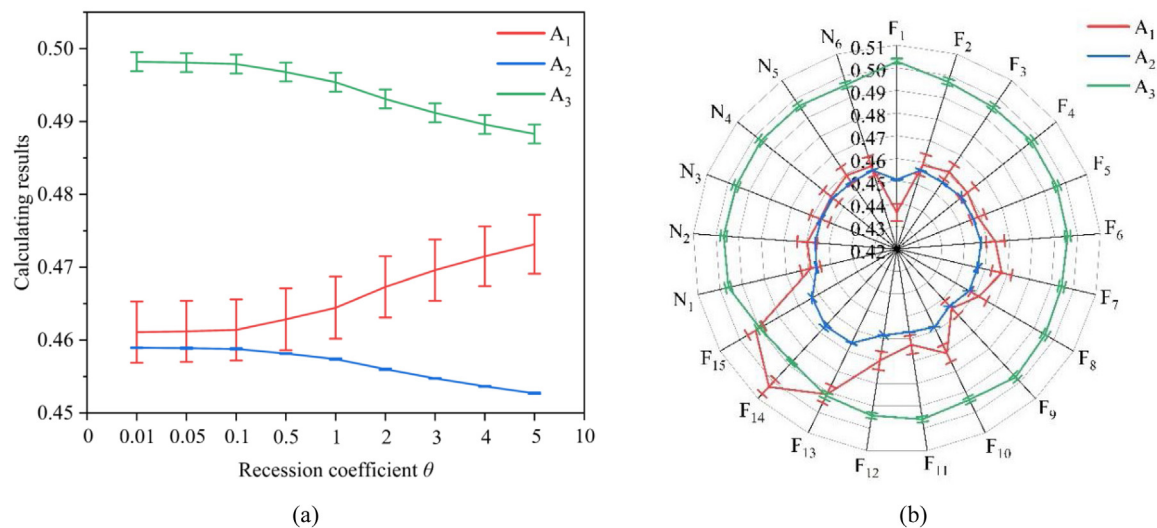


Fig. 7. (a) Sensitive analysis of recession coefficient; (b) Sensitive analysis of criterion weight.

material selection of 3D printed composites, some limitations remain. (1) only three alternatives of 3D printed composites are considered and analyzed in this study. A study with more printing parameters, e.g., stacking orders, matrix material types, and fiber types, should be carried out to explore the effects of multiple printing parameters on the material selection results. (2) In the process of material selection, mechanical criteria are commonly obtained with experimental errors, which have not been fully considered in this study. An effective tool needs to be proposed to depict the experimental errors and probe the effects of experimental errors on the decision results. (3) This study can effectively solve the uncertain problems and hesitancy problems in the decision-making process. But the decision behaviors, e.g., prospect theory and regret theory, of decision-makers have not been further studied.

7. Conclusion

In this study, to address the material selection of 3D printed composites, a multistage decision-making approach with PIVHFS is proposed. The literature reviews are conducted to illustrate the research gaps of material selection issues, i.e., uncertain information issues in material selection, lacking systemic criterion model, and unexplored interaction and correlation effects. To settle the above gaps, the PIVHFS is presented to solve the uncertain and fuzzy problems in material selection, including hesitation information, incomplete divergence, and judgment. Based on the PIVHFS, the fuzzy integral Shapley value is generalized to the PIVHFS-based Shapley value. The PIVHFS-based Shapley value is further applied to quantify the effects of interactions among criteria and determine the interaction importance weights of each criterion. In addition, the PIVHFS-based TODIM-TOPSIS method is proposed by using the TODIM method to transform the dominance of alternative to the dominance of alternative and further consider the criterion correlation effects. In this way, the correlation and dominance effects of multiple criteria are considered to fill the gaps in the TOPSIS method and further rank the alternatives. A hierarchical multiple criterion model is constructed, comprehensively considering the stability of the hygrothermal aging environment along with mechanical, economic, environmental, and social aspects, to describe the attributes of various aspects of the composites. As a case study, three 3D printed continuous glass fiber reinforced composite material alternatives manufactured with different printing sequences, printing directions, and matrix/fiber types, are introduced in the background of automotive chassis to validate the proposed MCDM approach. A decision matrix construction method incorporating mechanical tests and expert evaluation is also

presented. Finally, the fuzzy TODIM method, TOPSIS method, AHP method, and fuzzy AHP method are introduced to conduct a comparative analysis, and the sensitive analysis of recession coefficient and criterion weights is also presented.

The decision results show that mechanical properties (F₁₃–F₁₅), material cost (F₁), and light-weight level (F₉) exhibit higher importance in the material selection. The contribution of stability criteria is relative equilibrium, and its overall importance level is second only to the mechanical properties. The SS-SCGFLs is selected as the optimal alternative, and CS-SCGFLs is the worst. It illustrates that the addition of continuous fibers may not improve the dominance of 3D printed composites. However, by applying different printing strategies, the dominance of composites can be effectively improved. Therefore, for 3D printed composites in harsh environments, a higher priority can be achieved by optimizing the design of printing strategies. This study provides a hierarchical multiple criterion model and a trustworthy MCDM approach for the selection of 3D printed composites in harsh environments. The comparative analysis illustrates that the proposed approach presents a higher stable, accurate, and feasible characteristic. This study has extensive managerial implications and also serves as a resource for upcoming research on adjacent topics.

Eventually, future studies will be devoted to addressing the limitations that remain in this study. 3D printed composites with more printing parameters and alternatives will be further studied, and the effects of experimental errors in mechanical properties need to be considered. In addition, the effects of decision behavior in material selection will also be explored.

CRediT authorship contribution statement

Guoquan Xie: Methodology, Software, Validation, Formal analysis, Writing – original draft, Visualization. **Kui Wang:** Investigation, Resources, Writing – review & editing, Supervision, Funding acquisition. **Xuan Wu:** Methodology, Formal analysis, Writing – original draft. **Jin Wang:** Data curation, Visualization. **Tao Li:** Methodology, Software. **Yong Peng:** Funding acquisition, Supervision, Resources. **Honghao Zhang:** Conceptualization, Writing – review & editing, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.engappai.2023.106483>.

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